AI BUILDER CUP · SUBMISSION

QuantarisAI

AI manufacturing intelligence: from one sentence to a plant that runs clean

Team details

1. **Team name:** [add your team name]
2. **Team leader name:** Krishna Tandon
3. **Problem statement:**

Build an AI-powered solution that addresses challenges related to efficiency, quality, reliability, supply and demand, or sustainability across manufacturing and industrial operations. Solutions should demonstrate how AI can transform operational data into meaningful insights, recommendations, or actions.

Most owners start with a material, not a plan

100 kg of coconut husk every day. Forty cows. A pineapple farm.
The owner has the raw material and the will, but not the answers.

Today they pay consultants, or guess. Pollution, wasted water and the wrong site are discovered after the money is spent.



What should I build?

Which product, which steps, which machines



Where should I build it?

Which area, which sources, what it costs



How do I run it clean?

Water, smoke and waste: what can be reused

QuantarisAI plans it, places it, then helps run it clean

Plan

Describe a factory in English or Hindi. Get a researched blueprint: steps, machines, a 3D plant with raw material arriving and product leaving, and a farm beside the gate.

Place

Find the best areas in your city with sources and a map, the nearest suppliers, indicative machine prices and land options.

Run clean

See how water, smoke and waste can be reused, then load your own plant records: AI finds the leak, the weak machine, the quality drift.

Honest by design. Every claim is labelled fact, assumption, calculation, idea to validate, simulation or unknown. Where data is simulated or a figure is unknown, the app says so.

Opportunities: why this is different and why it works

How is it different?

- Starts before the factory exists, then follows it into operations
- Researches real places and suppliers with Google Search, with source links
- Shows the evidence: fact, assumption, calculation, or unknown

How does it solve the problem?

- Efficiency: bottleneck and OEE
- Quality: photo inspection, defect drift
- Reliability: failing machines found early
- Supply and demand: sources, 13-week forecast
- Sustainability: reuse loops, smoke and water with AI

Our USP

- One tool from idea to operations, in plain words
- English and Hindi, voice and text
- Works on the owner's own CSV, not only demos
- Gemini plus Google Search, with guardrails

What the solution offers

Chat-first design

English and Hindi, voice or text, edit the plant by speaking

3D plant

Raw material in, product out, people, farm, smoke

Best location

Ranked real areas with sources and map pins

Sourcing and costs

Nearest suppliers, indicative machine and land prices

Reuse and save

Water, smoke and heat, waste: Today, Fix, Result

Plant AI

Maintenance, OEE, quality drift, demand, smoke and water

Your data

Upload a CSV: use per unit, gap to your best, reuse levers

Copilot + vision

Copilot quotes the numbers; Gemini vision checks a photo

Process flow: from one sentence to daily operations

1

Describe

Text or voice, English or Hindi. Photos for an existing plant.

2

AI drafts

Steps, equipment, waste, raw material in and product out.

3

Research

Google Search: sources, costs, best locations, good practice.

4

Visualise

3D plant, material flow, farm. Edit by voice.

5

Improve

Reuse loops. Pollution with and without AI.

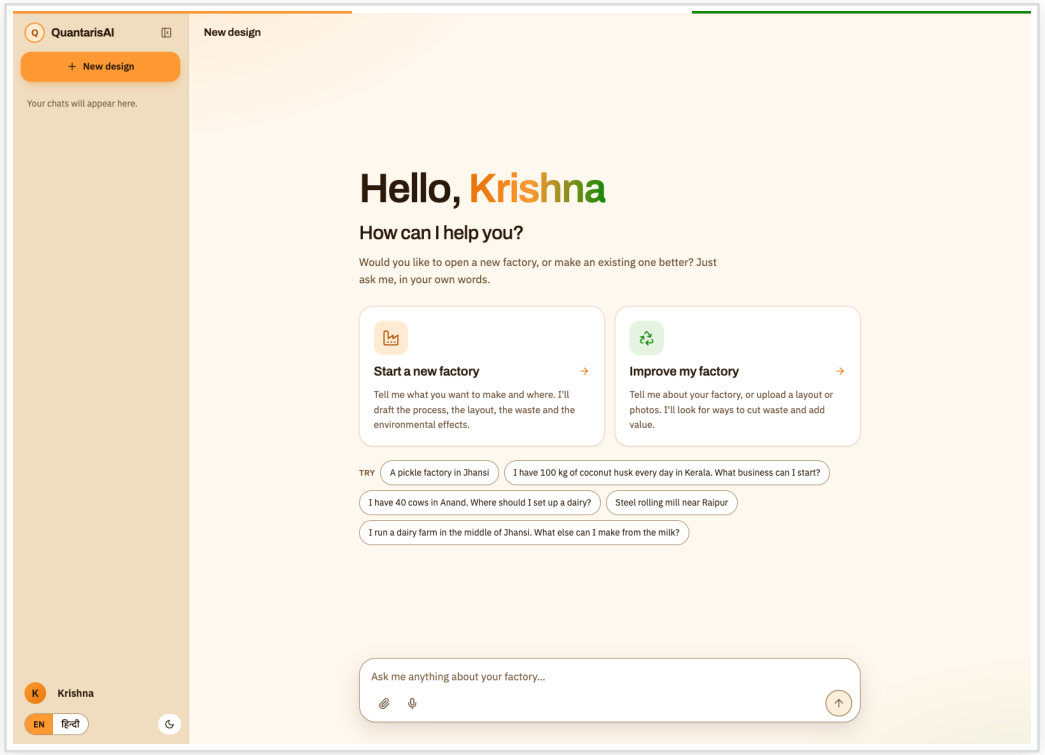
6

Operate

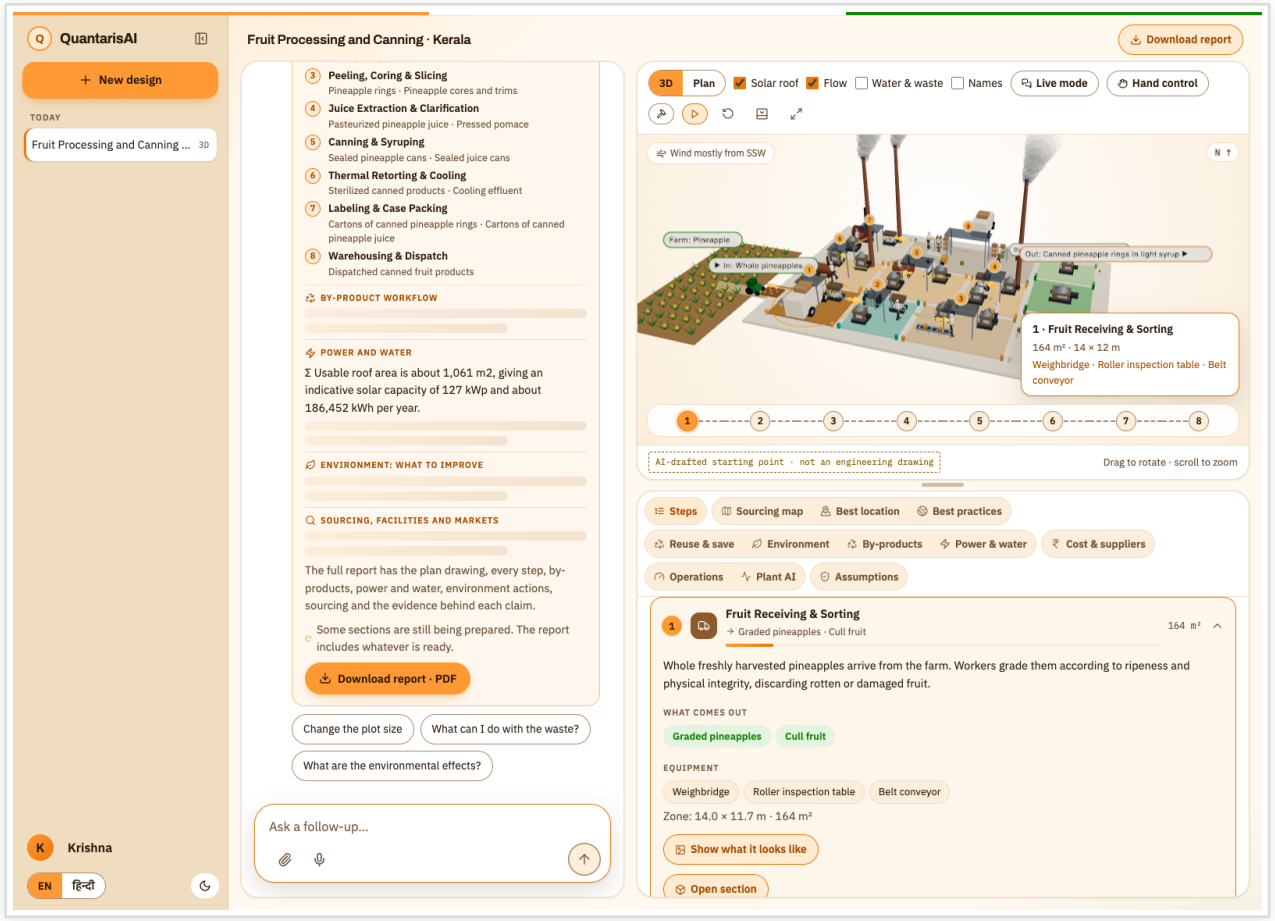
Your data in, insights and actions out. Ask the AI.

Where AI acts: Gemini designs and explains; Google Search grounds the facts; Gemini vision checks photos; plain statistics, not a model's guess, compute every trend, forecast and balance.

Screens: chat first, plant and tabs beside it

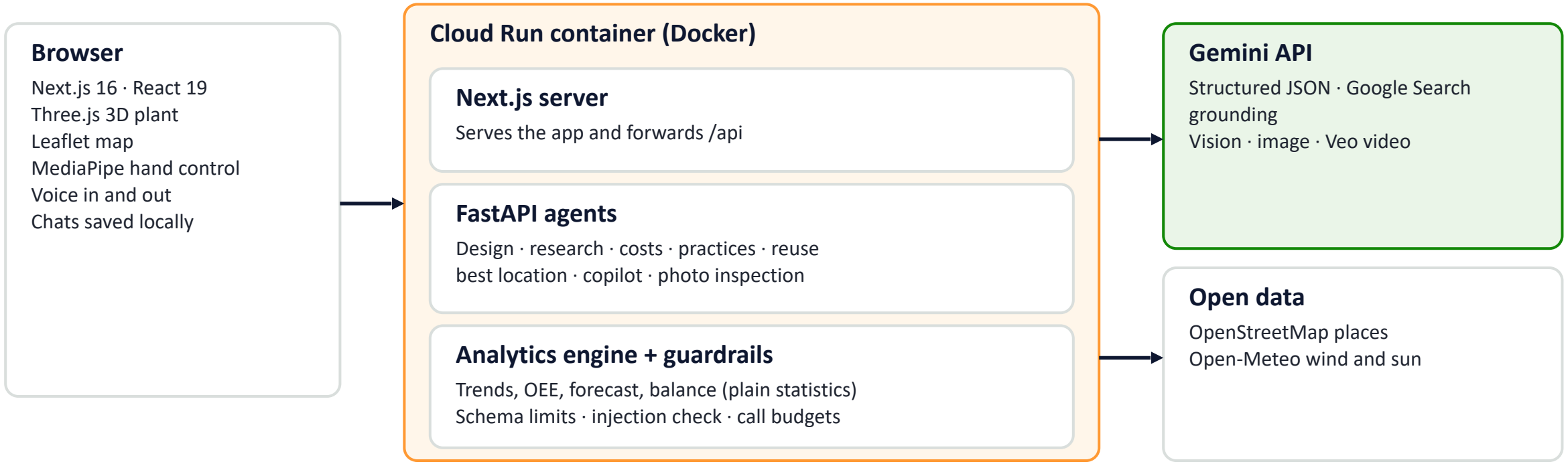


Landing: two clear choices, in English or Hindi, light or dark



Workspace: chat on the left, 3D plant above, grouped tabs below

Architecture: one container on Cloud Run, Gemini behind it



Planned next (not built yet)

Firebase Auth + Firestore (Google login, saved chats) · BigQuery + Pub/Sub + Dataflow (plant data pipeline) · Vertex AI (Gemini on Vertex, forecasting) · Cloud Storage (files and photos) · Secret Manager holds the Gemini key today

Technologies: used today and planned

Used today

Gemini API: structured JSON, Google Search grounding, vision, image generation, Veo video

FastAPI + Pydantic: agents, bounded outputs, call budgets

Next.js, React, Three.js: chat app, 3D plant, Leaflet map, MediaPipe

Docker, Cloud Run, Secret Manager: deployment-ready, key kept out of the image

OpenStreetMap, Open-Meteo: places, wind and sunshine

Planned

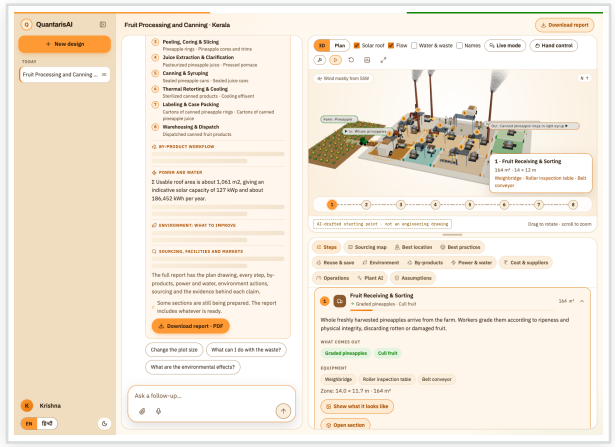
- Vertex AI (Gemini on Vertex, forecasting)
- BigQuery (plant data)
- Pub/Sub and Dataflow (streams)
- Cloud Storage (files, photos)
- Firebase Auth and Firestore
- Vision AI for inspection at scale

Estimated implementation cost: what drives it

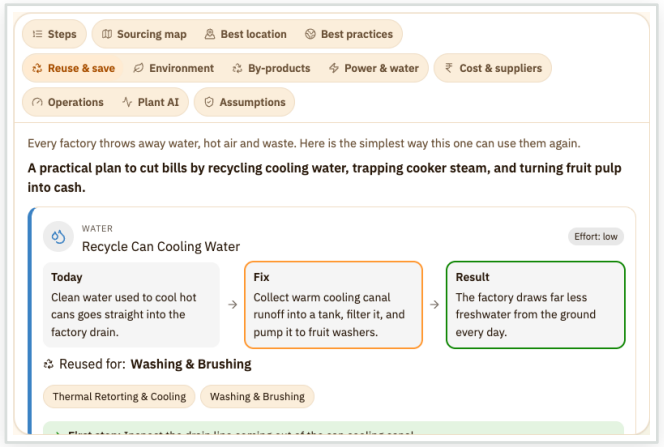
Item	What drives the cost	Today
Gemini calls	About six web requests build one full plan, each one to two model calls; photos, video and the copilot are extra and on demand	Not yet measured in rupees
Cloud Run	One 1 GB container; can scale to zero when idle, or keep one instance warm for a demo	Deployment-ready
Storage and database	None on the server: chats stay in the browser	No cost yet
Cost control	Per-hour call limits per instance; max instances capped at deploy time	In place
Next phase	BigQuery, Firebase and Vertex AI bring usage-based costs when real plants connect	To be estimated

We have not turned usage into a rupee figure: model pricing changes, and we would rather say so than invent one.

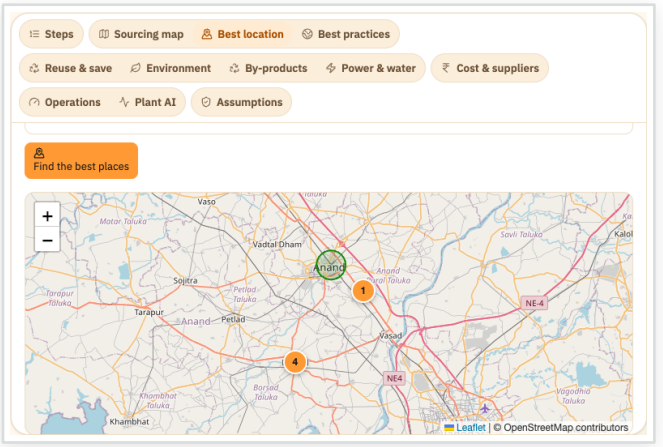
Snapshots of the working prototype



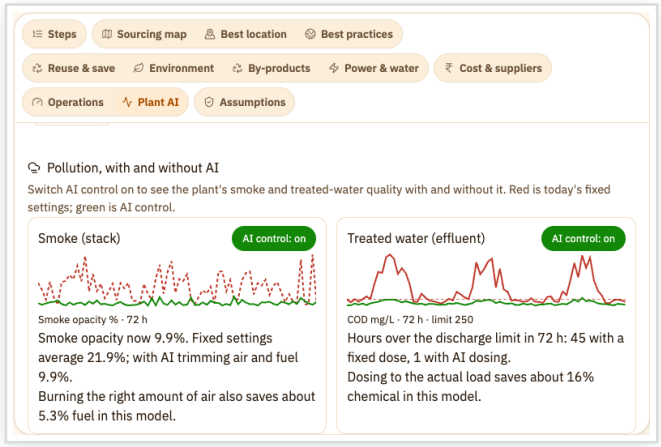
Pineapple plant: farm in, cans out



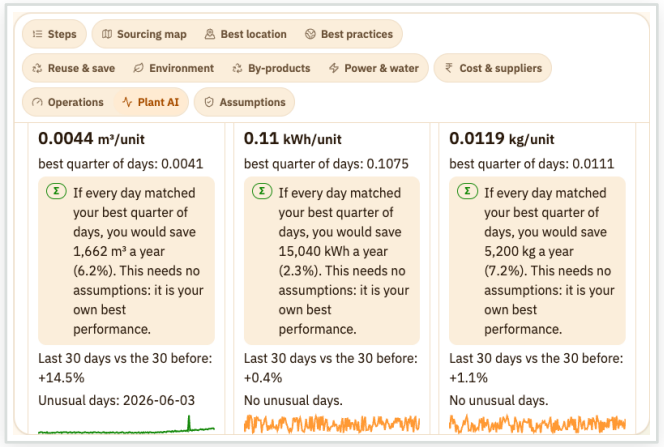
Reuse and save: Today, Fix, Result



Best location: ranked areas on the map



Smoke and water, with and without AI



Your data: per-unit use, gap, trend

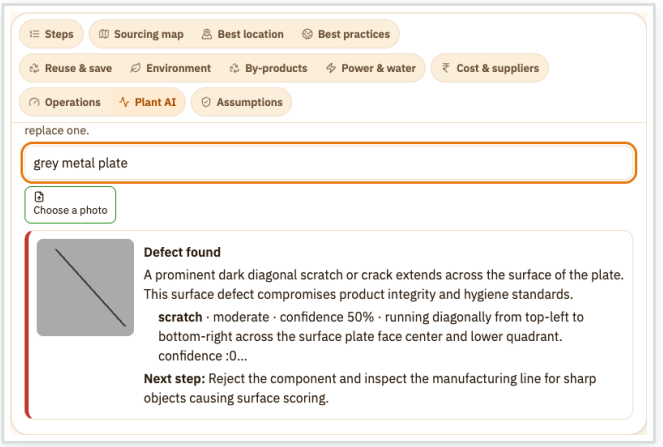
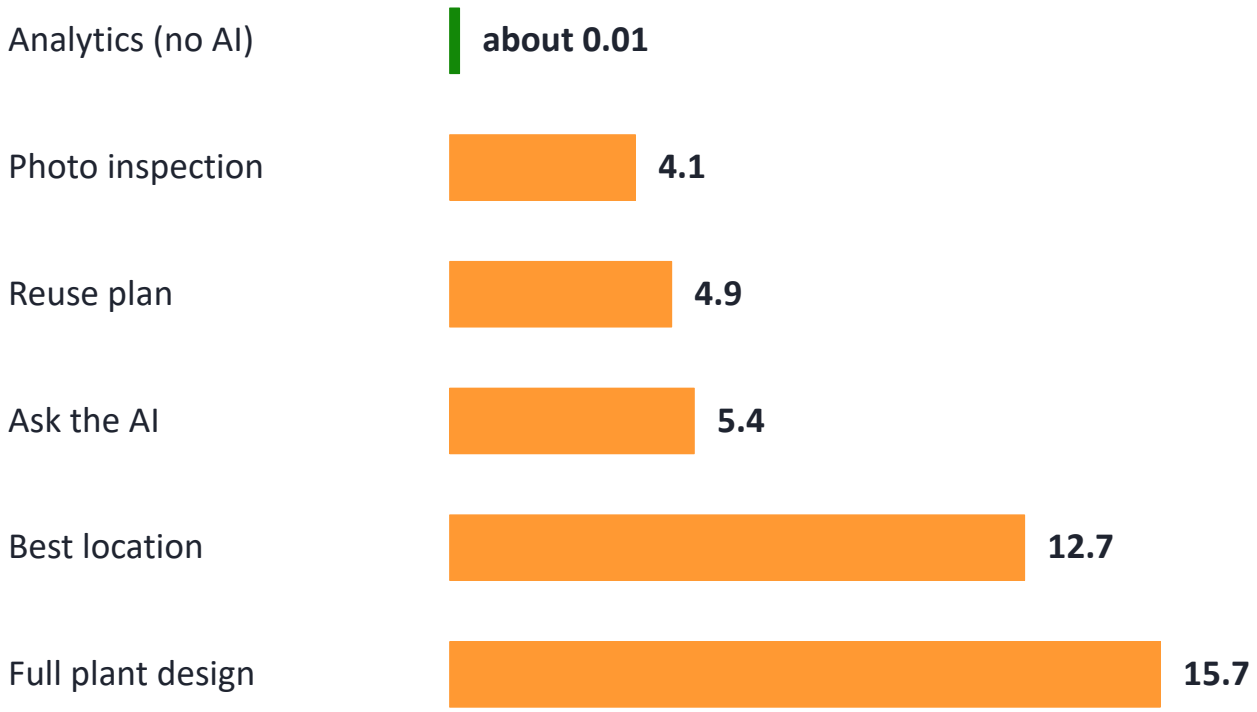


Photo inspection with Gemini vision

Prototype performance: measured, not promised

Response time in seconds (live Gemini)



Measured 18 Oct 2026 on a laptop through the web proxy; AI timings are single live runs, analytics are the median of five. The plant data behind the forecast is simulated.

229
backend tests passing

92
frontend tests passing

5 to 13%
forecast back-test error on simulated plants

+14.5%
water per unit trend found in the dairy sample (planted problem)

Photo inspection of a scratched test plate: verdict fail, defect scratch, 98% confidence.

Additional details: what comes next and what is limited today

Build next

- Google login and saved chats (Firebase)
- Plant data pipeline: BigQuery, Pub/Sub, Dataflow
- Forecasting and anomaly models on Vertex AI
- Upload sensor and sales files, Excel as well as CSV
- Exact map pins with the Google Maps API
- Realistic 3D models (Blender)
- Measure inspection accuracy on real products

Honest limits today

- Machine, production, quality and demand numbers are simulated until a CSV is loaded
- Prices are indicative web figures, not quotes
- Location ranking is AI judgement; some places are not pinned
- People, animals and machines are stylised
- Hand control not tested with a real hand
- Nothing is stored on the server

Links

GitHub public repository

https://github.com/sleepyowl1208/GoogleHackathon_QuantarisAI

Demo video (3 minutes)

[add the video link]

Final product link

[add the Cloud Run URL after deploying]

The seven suggested directions: what is real today

Direction	What it does	Status
Predictive maintenance	Trend and time-to-limit per machine; schedule the fix	**Simulated data; CSV upload next**
Visual quality inspection	Gemini vision finds defects in your photo	**Real**
Production optimization	Bottleneck, OEE, capacity what-if	**Simulated data**
Supply-chain intelligence	Nearest sources, lead time, risks, order-by week	**Real research, simulated risk figures**
Demand forecasting	13-week range, back-tested error shown	**Simulated history**
Waste reduction	Reuse loops; water and waste balance from your CSV	**Real on your data**
Operational copilot	Voice and text; quotes only computed numbers	**Real**

Describe it. Place it. Run it clean.

QuantarisAI helps an owner with a raw material and a plan decide what to build, where, and how to cut water, smoke and waste, with the evidence shown and nothing faked.

github.com/sleepyowl1208/GoogleHackathon_QuantarisAI

Thank you